

Artificial
Intelligence



ХИЙМЭЛ ОЮУН УХААН – J.CS712

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КУ-ны салбар

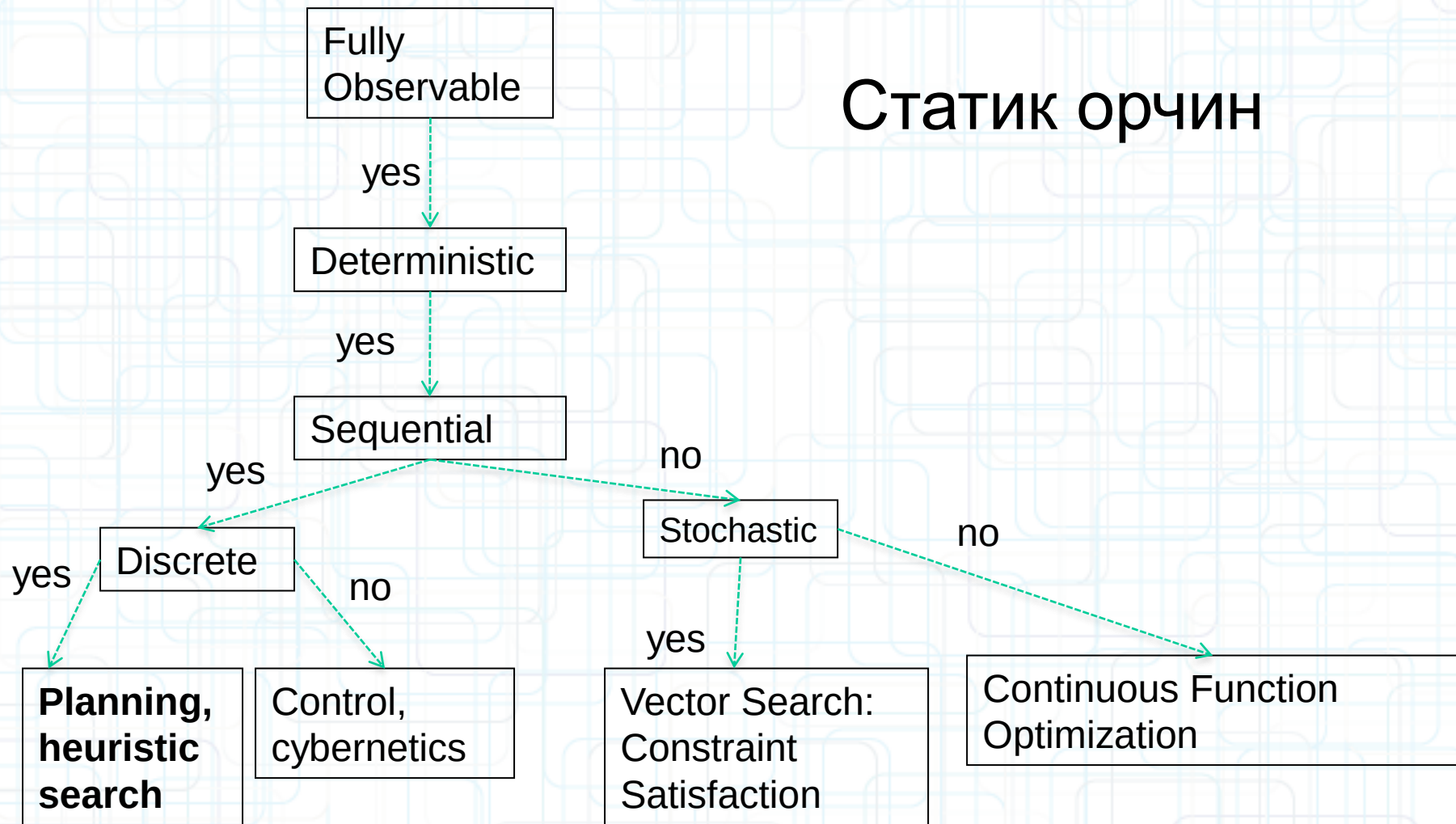
ШУТИС МХТС, 2021

Мэдээлэлгүй хайлт

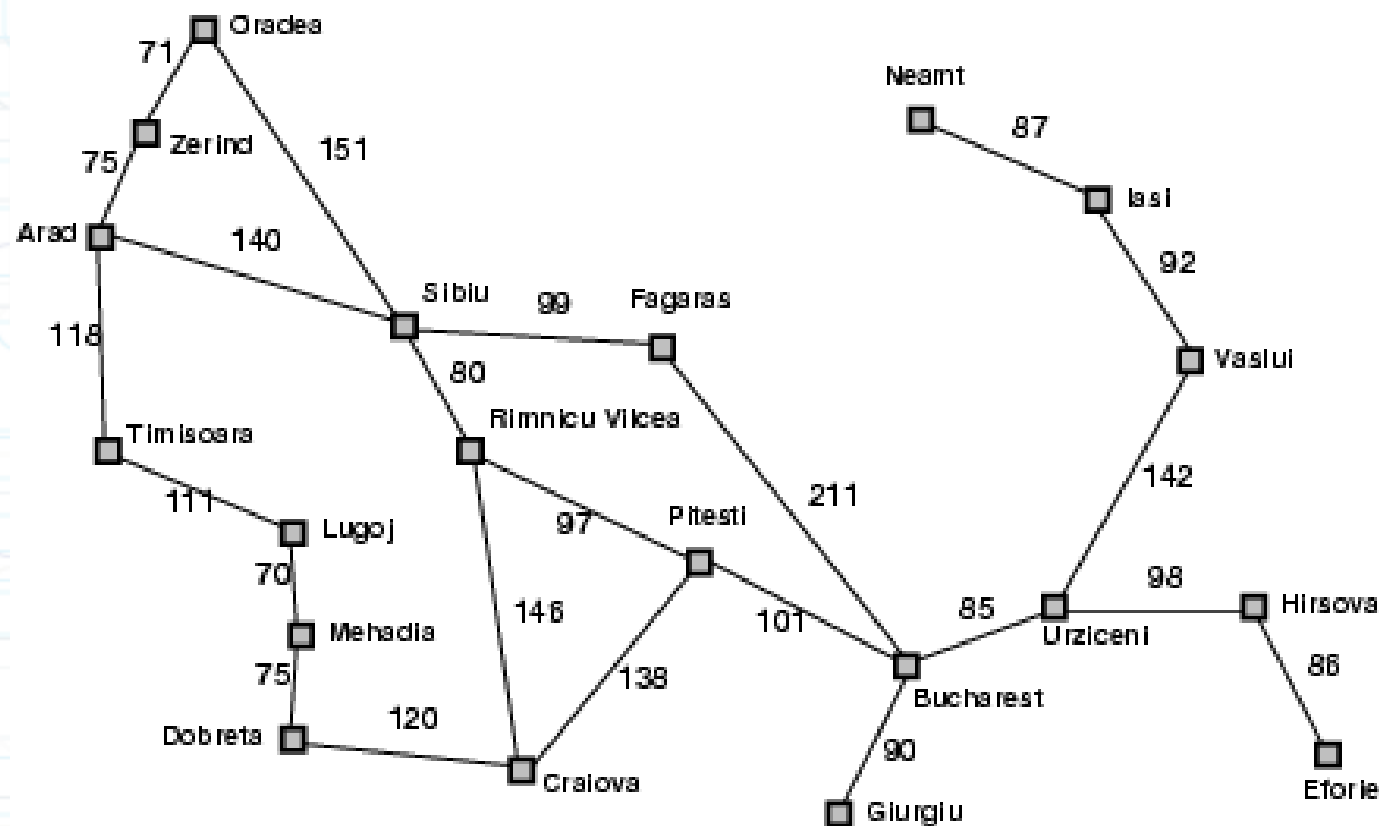
- Best-first search
- A^* search
- Heuristics
- Local search algorithms
- Hill-climbing search
- Simulated annealing search
- Local beam search

Мэдээлэлгүй хайлт

Статик орчин



Romania алхам өртөг km-ээр



Straight-line distance
to Bucharest

Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	176
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	10
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Greedy best-first хайлт

Evaluation function

$$f(n) = h(n) \text{ (heuristic)}$$

= estimate of cost from n to *goal*

e.g., $h_{SLD}(n)$ = straight-line distance from n to
Bucharest

Greedy best-first search expands the node that
appears to be closest to goal

Greedy best-first хайлт

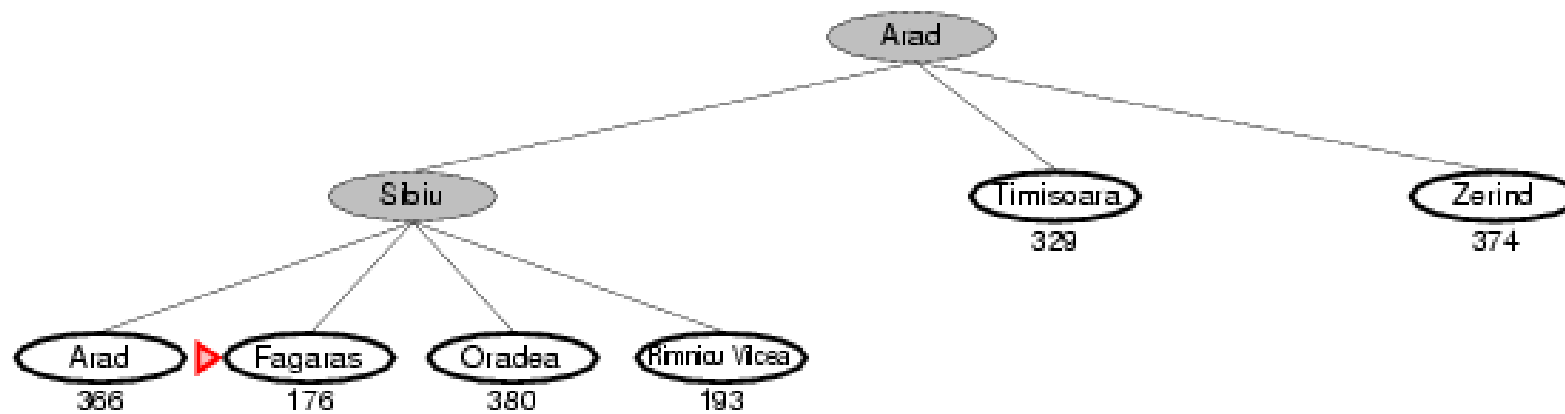


Arad
366

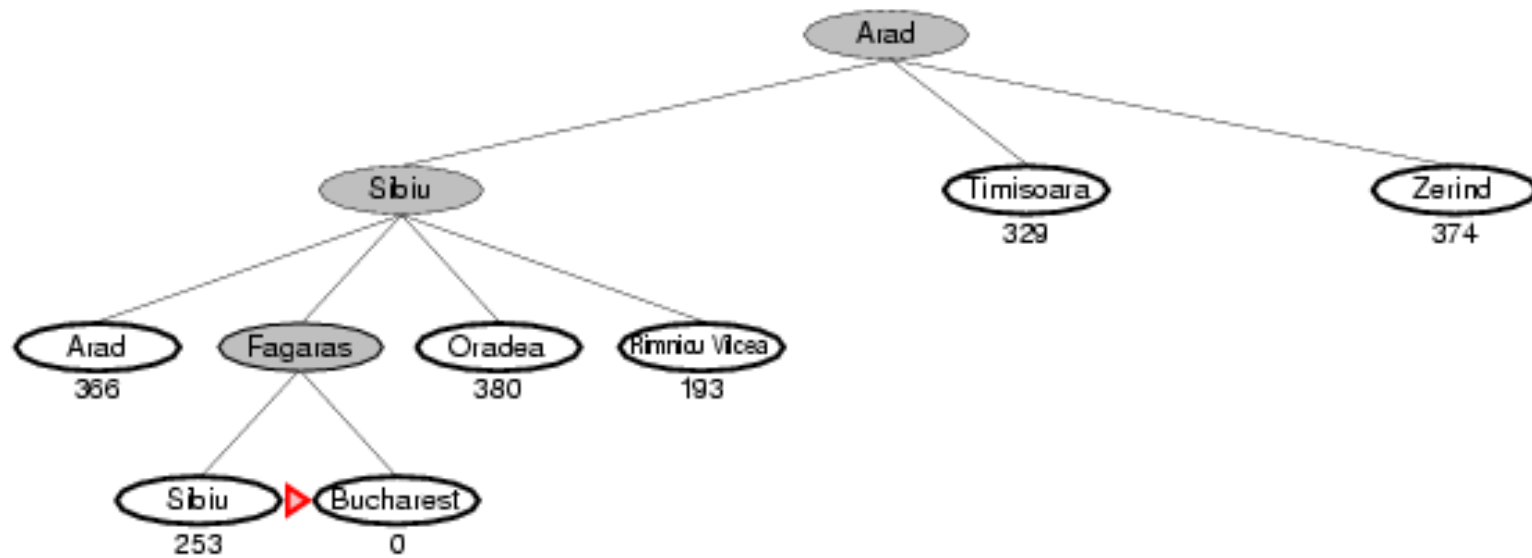
Greedy best-first хайлт



Greedy best-first хайлт



Greedy best-first хайлт



Greedy best-first хайлт

Complete? No – can get stuck in loops,
e.g. as Oradea as goal

IASI → Neamt → IASI → Neamt →

Time? $O(b^m)$, but a good heuristic can give
dramatic improvement

Space? $O(b^m)$ -- keeps all nodes in memory

Optimal? No

A* хайлт

Idea: avoid expanding paths that are already expensive.

Very important!

Evaluation function $f(n) = g(n) + h(n)$

$g(n)$ = cost so far to reach n

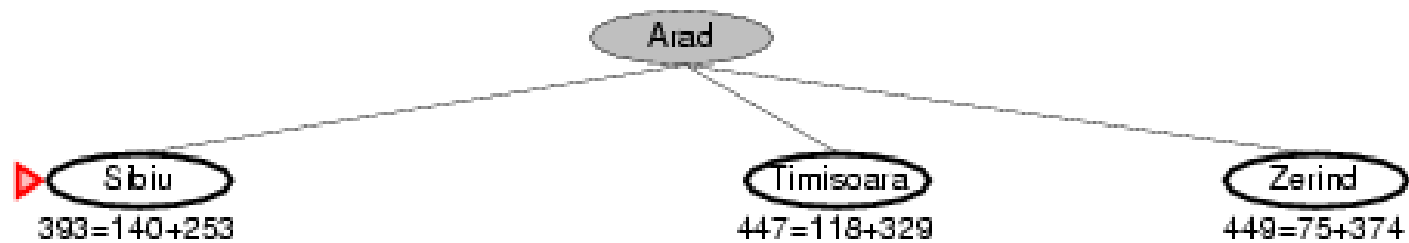
$h(n)$ = estimated cost from n to goal

$f(n)$ = estimated total cost of path through n to goal

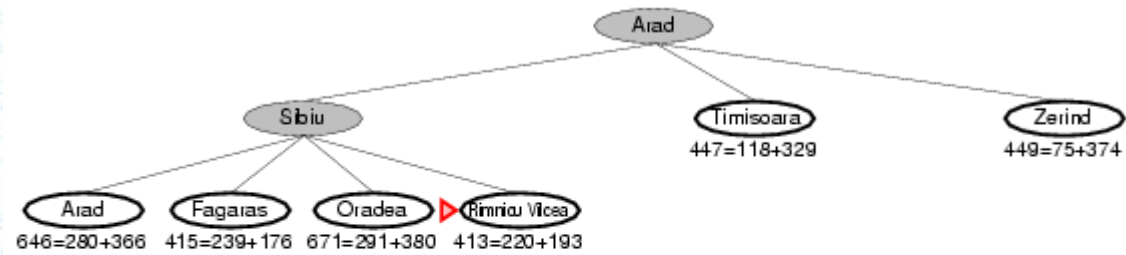
A* хайлт

 Arad
366=0+366

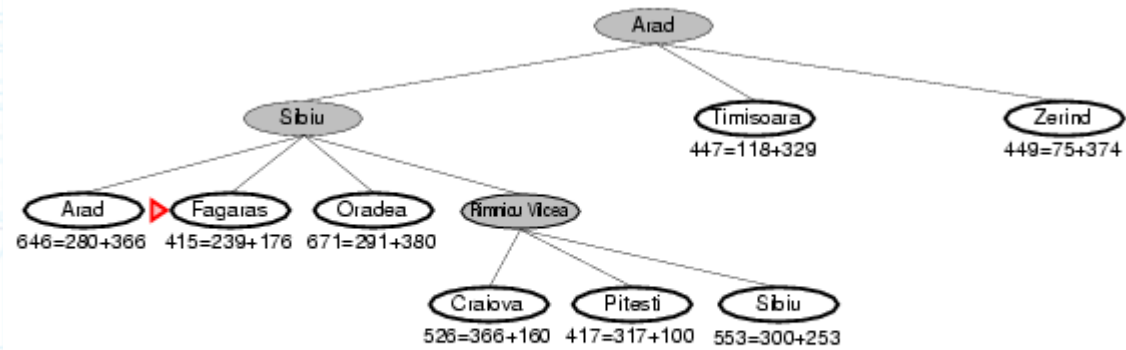
A* хайлт



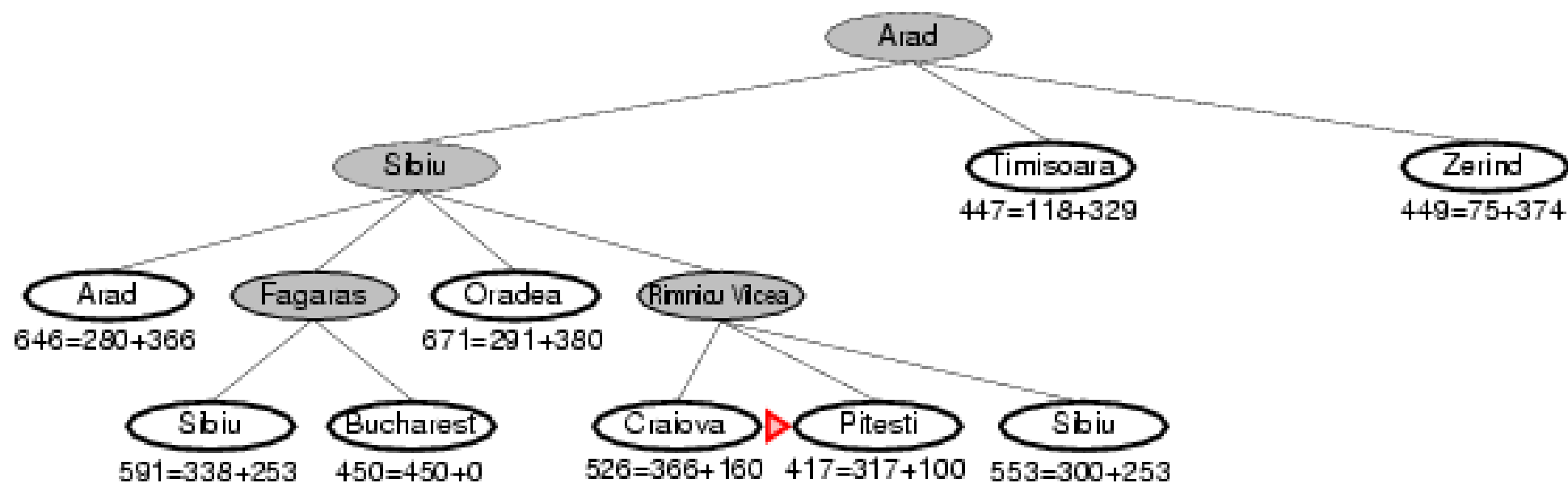
A* хайлт



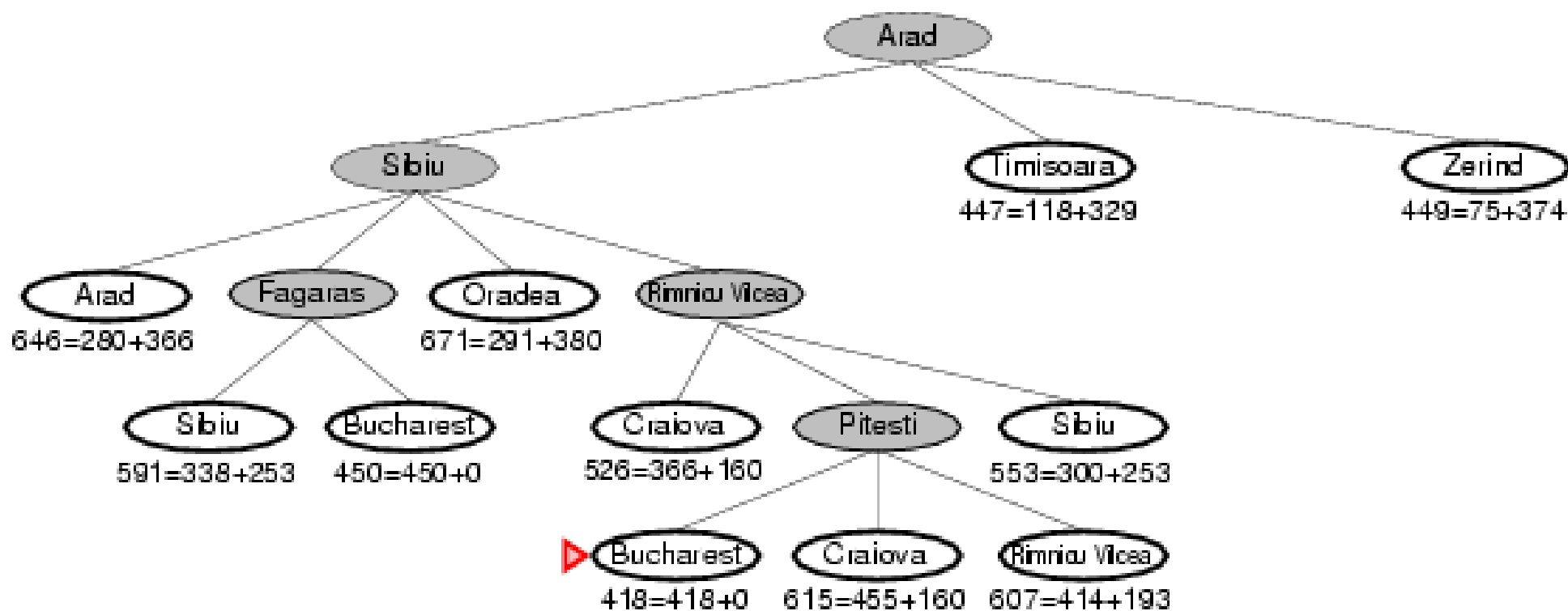
A* хайлт



A* хайлт



A* хайлт



A* хайлт

Complete? Yes (unless there are infinitely many nodes with $f \leq f(G)$)

Time? Exponential

Space? Keeps all nodes in memory

Optimal? Yes

A* хайлт

Complete? Yes (unless there are infinitely many nodes with $f \leq f(G)$)

Time? Exponential

Space? Keeps all nodes in memory

Optimal? Yes

A* хайлт

E.g., for the 8-puzzle:

$h_1(n)$ = number of misplaced tiles

$h_2(n)$ = total Manhattan distance

(i.e., no. of squares from desired location of each tile)

7	2	4
5		6
8	3	1

Start State

	1	2
3	4	5
6	7	8

Goal State

$$\frac{h_1(S)}{h_2(S)} = ?$$

A* хайлт

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(i.e., no. of squares from desired location of each tile)

7	2	4
5		6
8	3	1

Start State

	1	2
3	4	5
6	7	8

Goal State

$$\underline{h_1(S)} = ? \quad 8$$

$$\underline{h_2(S)} = ? \quad 3+1+2+2+2+3+3+2 = 18$$

A* хайлт

- If $h_2(n) \geq h_1(n)$ for all n (both admissible) then h_2 dominates h_1 .
- h_2 is better for search
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- Typical search costs (average number of nodes expanded):
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- $d=12$ IDS = 3,644,035 nodes
 $A^*(h_1) = 227$ nodes
 $A^*(h_2) = 73$ nodes
- $d=24$ IDS = too many nodes
 $A^*(h_1) = 39,135$ nodes
 $A^*(h_2) = 1,641$ nodes
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